Assignment Solution

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Introduction to the IS-LM Model

- The **IS-LM model** is a fundamental macroeconomic tool used to analyze the short-run fluctuations in an economy.
- It depicts the interaction between two markets:
  1. **Goods Market (IS Curve):** Investment and Saving.
  2. **Money Market (LM Curve):** Liquidity preference and Money supply.
- The intersection determines the general equilibrium for interest rates (r) and real output/income (Y).

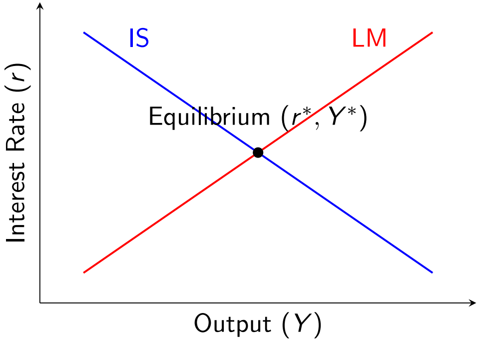
The IS Curve (Goods Market Equilibrium)

- **Definition:** Represents all combinations of interest rates (r) and output (Y) where the goods market is in equilibrium.
- **Equation:** $Y = C(Y - T) + I(r) + G$
- **Logic:**
  - As the interest rate rises ($r \uparrow$), the cost of borrowing increases.
  - Investment demand falls ($I \downarrow$).
  - Lower investment reduces aggregate demand and output ($Y \downarrow$).
- **Slope:** Downward sloping.

The LM Curve (Money Market Equilibrium)

- **Definition:** Represents all combinations of r and Y where the demand for real money balances equals the supply.
- **Equation:** $\frac{M}{P} = L(r, Y)$
- **Logic:**
  - As income rises ($Y \uparrow$), the demand for money for transactions increases.
  - With a fixed money supply (M), the interest rate must rise to restore equilibrium ($r \uparrow$).
- **Slope:** Upward sloping.

General Equilibrium Visualization



The point where IS intersects LM represents simultaneous equilibrium in both markets.

Fiscal Policy Analysis

- **Expansionary Fiscal Policy:** Increase in Govt. Spending (G) or Decrease in Taxes (T).
- **Effect:**
  - Shifts the **IS Curve** to the **Right**.
- **Result:**
  - Output increases ($Y \uparrow$).
  - Interest rates rise ($r \uparrow$).
- *Note:* The rise in r can cause "Crowding Out" of private investment.

Monetary Policy Analysis

- **Expansionary Monetary Policy:** Central Bank increases Money Supply (M).
- **Effect:**
  - Shifts the **LM Curve** to the **Right** (or Down).
- **Result:**
  - Interest rates fall ($r \downarrow$).
  - Lower rates stimulate investment, increasing output ($Y \uparrow$).

Summary of Shocks

Policy Event	**Curve Shift**	**Output (Y)**	**Interest (r)**
Govt Spending $\uparrow$	IS shifts Right	Increases	Increases
Taxes $\uparrow$	IS shifts Left	Decreases	Decreases
Money Supply $\uparrow$	LM shifts Right	Increases	Decreases
Money Demand $\uparrow$	LM shifts Left	Decreases	Increases